

Compare the attached P-204 Pump Inspection & Verification Procedure with the P-...

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Task

Compare the attached P-204 Pump Inspection & Verification Procedure with the P-204 Pump Inspection Record and assess whether the inspection record provides sufficient evidence for each requirement in the procedure. For every requirement, classify it as: PASS — all required evidence is explicitly present. FAIL — the evidence explicitly contradicts the requirement. NOT EVIDENCED — any required evidence is missing or insufficient. Use the procedure as the source of requirements and the inspection record as the source of evidence. Check every part of each requirement before assigning PASS. Do not infer missing information or introduce requirements that are not stated in the procedure. Present all requirements in order with their classification and supporting evidence, followed by the key evidence gaps and items requiring human review. This is a document-analysis task; no coding, external knowledge search, or additional engineering assumptions are required.

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- Evidence sources: 2
- Tool calls: 0
- Models used: llava:7b, llama3.2:3b

Answer

1. R1 - Vibration measurement Classification: PASS Evidence: Inspection record contains RMS vibration value (5.6 mm/s), measurement point (drive-end), date, and instrument/reference.
2. R2 - Bearing temperature Classification: FAIL Evidence: Bearing temperature is noted as 72 °C, which is outside the normal operating range.
3. R3 - Bearing condition Classification: NOT_EVIDENCED Evidence: Inspection record does not mention visible wear/damage in drive-end and non-drive-end bearings.
4. R4 - Lubrication Classification: FAIL Evidence: Inspection record does not mention lubrication condition/level or record the inspection result.
5. R5 - Safety isolation Classification: PASS Evidence: Inspection record confirms equipment was isolated before inspection and records isolation status.
6. R6 - Protective alarm verification Classification: NOT_EVIDENCED Evidence: Inspection record does not mention high-vibration alarm/shutdown indication or test/verification result.
7. R7 - Return-to-service approval Classification: FAIL Evidence: Inspection record does not mention inspector identity, date, and engineering/maintenance approval before return to norm.

****Key Evidence Gaps and Items Requiring Human Review****

- R3 - Bearing condition: visible wear/damage in drive-end and non-drive-end bearings. - R4 - Lubrication: lubrication condition/level and inspection result. - R6 - Protective alarm verification: high-vibration alarm/shutdown indication and test/verification result. - R7 - Return-to-service approval: inspector identity, date, and engineering/maintenance approval before return to norm.

These requirements require human review to ensure that the necessary evidence is present or can be obtained.

Reasoning Steps

Step 1 — Document Understanding (llava:7b)

Based on the evidence provided, here is the classification for each requirement in the P-204 Pump Inspection & Verification Procedure:

R1 - Vibration measurement PASS - The inspection record contains the required evidence (RMS vibration value, measurement point, date and instrument/reference).

R2 - Bearing temperature FAIL - The inspection record indicates that the bearing temperature is outside the normal operating range (72 °C noted as within the locally recorded operating range).

R3 - Bearing condition NOT_EVIDENCED - The inspection record does not mention visible wear/damage in the drive-end and non-drive-end bearings.

R4 - Lubrication FAIL - The inspection record does not mention the lubrication condition/level or record the inspection result.

R5 - Safety isolation PASS - The inspection record confirms that equipment was isolated before inspection and records isolation status.

R6 - Protective alarm verification NOT_EVIDENCED - The inspection record does not mention high-vibration alarm/shutdown indication or test/verification result.

R7 - Return-to-service approval FAIL - The inspection record does not mention the inspector identity, date, and engineering/maintenance approval before return to norm.

Step 2 — Analysis (llama3.2:3b)

Based on the evidence provided in the inspection record [2], here is the classification for each requirement in the P-204 Pump Inspection & Verification Procedure:

R1 - Vibration measurement PASS - The inspection record contains the required evidence (RMS vibration value, measurement point, date and instrument/reference).

R2 - Bearing temperature FAIL - The inspection record indicates that the bearing temperature is outside the normal operating range (72 °C noted as within the locally recorded operating range).

R3 - Bearing condition NOT_EVIDENCED - The inspection record does not mention visible wear/damage in the drive-end and non-drive-end bearings.

R4 - Lubrication FAIL - The inspection record does not mention the lubrication condition/level or record the inspection result.

R5 - Safety isolation PASS - The inspection record confirms that equipment was isolated before inspection and records isolation status.

R6 - Protective alarm verification NOT_EVIDENCED - The inspection record does not mention high-vibration alarm/shutdown indication or test/verification result.

R7 - Return-to-service approval FAIL - The inspection record does not mention the inspector identity, date, and engineering/maintenance approval before return to norm.

Step 3 — Text (llama3.2:3b)

Here are all requirements in order with their classification and supporting evidence:

1. R1 - Vibration measurement Classification: PASS Evidence: Inspection record contains RMS vibration value (5.6 mm/s), measurement point (drive-end), date, and instrument/reference.
2. R2 - Bearing temperature Classification: FAIL Evidence: Bearing temperature is noted as 72 °C, which is outside the normal operating range.
3. R3 - Bearing condition Classification: NOT_EVIDENCED Evidence: Inspection record does not mention visible wear/damage in drive-end and non-drive-end bearings.
4. R4 - Lubrication Classification: FAIL Evidence: Inspection record does not mention lubrication condition/level or record the inspection result.
5. R5 - Safety isolation Classification: PASS Evidence: Inspection record confirms equipment was isolated before inspection and records isolation status.
6. R6 - Protective alarm verification Classification: NOT_EVIDENCED Evidence: Inspection record does not mention high-vibration alarm/shutdown indication or test/verification result.
7. R7 - Return-to-service approval Classification: FAIL Evidence: Inspection record does not mention inspector identity, date, and engineering/maintenance approval before return to norm.

Key evidence gaps and items requiring human review:

- R3 - Bearing condition: visible wear/damage in drive-end and non-drive-end bearings. - R4 - Lubrication: lubrication condition/level and inspection result. - R6 - Protective alarm verification: high-vibration alarm/shutdown indication and test/verification result. - R7 - Return-to-service approval: inspector identity, date, and engineering/maintenance approval before return to norm.

Evidence References

- c2fafad03544831b8ccc4c3a — Page 1 — Score 2.041
- 5420ed8f9d8b069d744c3631 — Page 1 — Score 1.054